Experiment - 02

EE2501

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Task 1

Aim

Measure and analyze the $I-V$ characteristics of a diode (1N4007) in forward and reverse bias using series resistances of 100 Ω, 220 Ω, and 4.7 kΩ. Find reverse-bias saturation current I_s, thermal voltage V_T, and ideality factor n by curve fitting with the Shockley equation.

Equipment and Components used

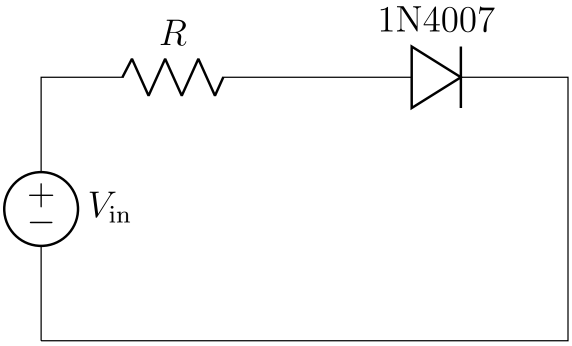
- Multimeter for measuring V_d and R
- DC Programmable Power Supply (Wanptek WPS305H) from Tinkerer's Lab
- Resistor of 100 Ω
- Resistor of 220 Ω
- Resistor of 4.7 kΩ
- 1N4007 diode
- Breadboard and jumpers

Procedure

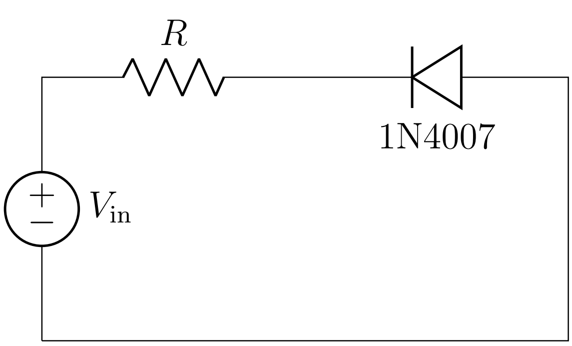
1. **Setup the circuit** Connect the DC source $V_{\text{in}} \to$ series resistor $R \to$ diode $D \to$ GND.
2. **Forward-bias sweep.** With diode anode towards V_{in}, slowly increase V_{in}.
   - At each step record V_{diode}, compute $V_{\text{resistor}} = V_{\text{in}}$ - V_{diode}
   $$I = V_{\text{res}}/R.$$
   - Repeat for $R = 100\ \Omega,\ 220\ \Omega,\ 4.7\ \text{k}\Omega$.
3. **Reverse-bias sweep.** Reverse the diode (cathode to V_{in}).
   - Sweep $|V_R|$ using small steps
   - Record V_{in}, V_{res}, V_{diode}, and compute I
4. **Curve fitting (Shockley).** Use the experimental data to fit to the below equation
   $$I \;=\; I_s\left(\exp\frac{V_{\text{diode}}}{nV_T} - 1\right).$$
5. **Visualization.** Plot $I-V$ curves for the 3 cases of R, overlay best-fit Shockley curve

Stage 1 : Circuit Design and Setup

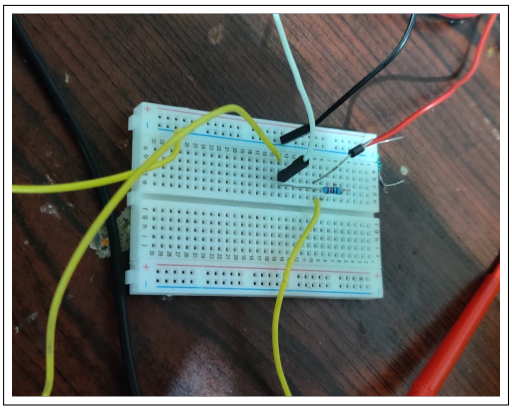
The diode is placed in series with a resistor and driven by a programmable DC power-supply. Voltage is measured across the diode to compute the branch current.



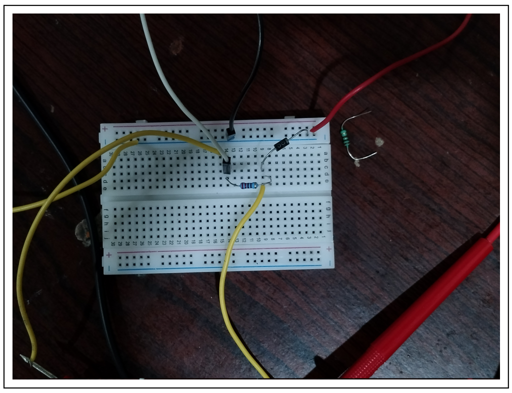
Forward bias



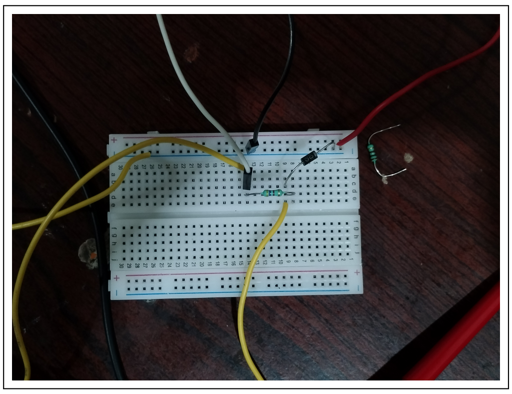
Reverse bias



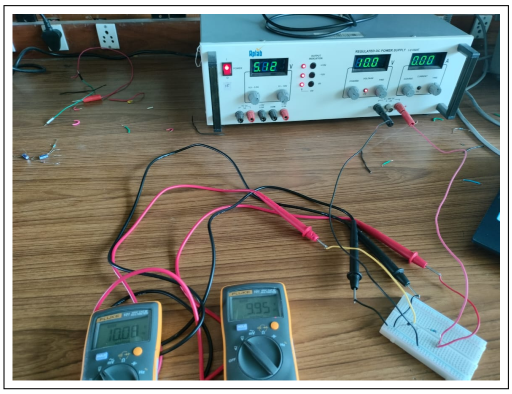
Breadboard implementation with 100 Ω



Breadboard implementation with 220 Ω



Breadboard implementation with 4700 Ω



Experimental Setup to take measurements

Observations

We have recorded the entire data in excel, as attached in the next page.

R = 100			
Vin	V_diode	V_r	i_r (mA)
-0.146	-0.145	-0.001	-0.01
-0.211	-0.209	-0.002	-0.02
-0.298	-0.294	-0.004	-0.04
-0.429	-0.425	-0.004	-0.04
-0.536	-0.531	-0.005	-0.05
-0.586	-0.58	-0.006	-0.06
-0.736	-0.729	-0.007	-0.07
-0.826	-0.817	-0.009	-0.09
-0.948	-0.939	-0.009	-0.09
-1.01	-0.999	-0.011	-0.11
-1.528	-1.512	-0.016	-0.16
-2.099	-2.076	-0.023	-0.23
-2.53	-2.504	-0.026	-0.26
-3.098	-3.065	-0.033	-0.33
-3.565	-3.528	-0.037	-0.37
-4.061	-4.017	-0.044	-0.44
-4.559	-4.512	-0.047	-0.47
-5.11	-5.054	-0.056	-0.56
-6.15	-6.08	-0.07	-0.7
-7.17	-7.09	-0.08	-0.8
-8.19	-8.1	-0.09	-0.9
-9.18	-9.08	-0.1	-1
0.1	0.097	0.003	0.03
0.2	0.2	0	0
0.3	0.3	0	0
0.4	0.4	0	0
0.5	0.495	0.005	0.05
0.6	0.563	0.037	0.37
0.61	0.566	0.044	0.44
0.62	0.573	0.047	0.47
0.63	0.578	0.052	0.52
0.64	0.581	0.059	0.59
0.65	0.586	0.064	0.64
0.66	0.59	0.07	0.7
0.67	0.593	0.077	0.77
0.68	0.597	0.083	0.83
0.69	0.6	0.09	0.9
0.7	0.603	0.097	0.97
0.71	0.607	0.103	1.03
0.72	0.61	0.11	1.1
0.73	0.612	0.118	1.18
0.74	0.615	0.125	1.25
0.75	0.619	0.131	1.31
0.76	0.62	0.14	1.4
0.77	0.623	0.147	1.47
0.78	0.625	0.155	1.55
0.79	0.627	0.163	1.63
0.8	0.629	0.171	1.71
0.81	0.632	0.178	1.78
0.82	0.634	0.186	1.86
0.83	0.636	0.194	1.94
0.84	0.638	0.202	2.02
0.85	0.639	0.211	2.11
0.86	0.641	0.219	2.19
0.87	0.642	0.228	2.28
0.88	0.645	0.235	2.35
0.89	0.647	0.243	2.43
0.9	0.648	0.252	2.52
1	0.661	0.339	3.39
1.1	0.672	0.428	4.28
1.2	0.682	0.518	5.18
1.3	0.689	0.611	6.11
1.4	0.696	0.704	7.04
1.5	0.702	0.798	7.98
2	0.723	1.277	12.77
2.5	0.737	1.763	17.63
3	0.748	2.252	22.52
3.5	0.756	2.744	27.44
4	0.763	3.237	32.37
4.5	0.769	3.731	37.31
5	0.774	4.226	42.26
5.5	0.779	4.721	47.21
6	0.782	5.218	52.18
6.5	0.785	5.715	57.15
7	0.788	6.212	62.12
7.5	0.792	6.708	67.08
8	0.795	7.205	72.05
8.5	0.797	7.703	77.03
9	0.799	8.201	82.01
9.5	0.8	8.7	87
10	0.802	9.198	91.98

R = 220			
Vin	V_diode	V_r	i_r
-0.114	-0.113	-0.001	-0.004545454545
-0.229	-0.226	-0.003	-0.01363636364
-0.322	-0.319	-0.003	-0.01363636364
-0.405	-0.4	-0.005	-0.02272727273
-0.551	-0.545	-0.006	-0.02727272727
-0.714	-0.709	-0.005	-0.02272727273
-0.842	-0.833	-0.009	-0.04090909091
-0.913	-0.903	-0.01	-0.04545454545
-1.017	-1.007	-0.01	-0.04545454545
-1.548	-1.529	-0.019	-0.08636363636
-2.026	-2.002	-0.024	-0.1090909091
-2.541	-2.515	-0.026	-0.1181818182
-3.011	-2.979	-0.032	-0.1454545455
-3.544	-3.506	-0.038	-0.1727272727
-4.009	-3.966	-0.043	-0.1954545455
-4.524	-4.475	-0.049	-0.2227272727
-5.077	-5.023	-0.054	-0.2454545455
-6.07	-6	-0.07	-0.3181818182
-7.02	-6.94	-0.08	-0.3636363636
-8.08	-8	-0.08	-0.3636363636
-9.04	-8.94	-0.1	-0.4545454545
-10.08	-9.95	-0.13	-0.5909090909
0.1	0.098	0.002	0.009090909091
0.2	0.2	0	0
0.4	0.4	0	0
0.6	0.543	0.057	0.2590909091
0.8	0.599	0.201	0.9136363636
1	0.628	0.372	1.690909091
1.5	0.666	0.834	3.790909091
2	0.689	1.311	5.959090909
2.5	0.703	1.797	8.168181818
3	0.714	2.286	10.39090909
3.5	0.723	2.777	12.62272727
4	0.73	3.27	14.86363636
4.5	0.736	3.764	17.10909091
5	0.74	4.26	19.36363636
5.5	0.746	4.754	21.60909091
6	0.75	5.25	23.86363636
7	0.758	6.242	28.37272727
8	0.765	7.235	32.88636364
9	0.769	8.231	37.41363636
10	0.774	9.226	41.93636364

R = 4700			
Vin	V_diode	V_r	i_r
-0.09	-0.09	0	0
-0.252	-0.247	-0.005	-0.001063829787
-0.336	-0.333	-0.003	0.0006382978723
-0.411	-0.407	-0.004	0.0008510638298
-0.523	-0.517	-0.006	-0.001276595745
-0.599	-0.593	-0.006	-0.001276595745
-0.782	-0.771	-0.011	-0.002340425532
-0.824	-0.816	-0.008	-0.00170212766
-0.89	-0.877	-0.013	-0.002765957447
-1.005	-0.992	-0.013	-0.002765957447
-1.114	-1.103	-0.011	-0.002340425532
-1.634	-1.616	-0.018	-0.003829787234
-2.09	-2.067	-0.023	-0.004893617021
-2.521	-2.492	-0.029	-0.006170212766
-3.067	-3.034	-0.033	-0.007021276596
-3.546	-3.507	-0.039	-0.00829787234
-4.034	-3.988	-0.046	-0.009787234043
-4.506	-4.455	-0.051	-0.01085106383
-5.057	-5.001	-0.056	-0.01191489362
-6.03	-5.96	-0.07	-0.01489361702
-7.06	-6.97	-0.09	-0.01914893617
-8.07	-7.98	-0.09	-0.01914893617
-9.05	-8.95	-0.1	-0.02127659574
-10.07	-9.93	-0.14	-0.02978723404
0.1	0.098	0.002	0.0004255319149
0.2	0.199	0.001	0.0002127659574
0.4	0.375	0.025	0.005319148936
0.6	0.445	0.155	0.0329787234
0.8	0.476	0.324	0.06893617021
1	0.495	0.505	0.1074468085
1.5	0.525	0.975	0.2074468085
2	0.545	1.455	0.3095744681
2.5	0.559	1.941	0.4129787234
3	0.57	2.43	0.5170212766
3.5	0.579	2.921	0.6214893617
4	0.586	3.414	0.7263829787
4.5	0.593	3.907	0.8312765957
5	0.599	4.401	0.9363829787
6	0.609	5.391	1.147021277
7	0.617	6.383	1.358085106
8	0.624	7.376	1.569361702
9	0.63	8.37	1.780851064
10	0.636	9.364	1.992340426

Curve Fitting to find η and I_s

The I-V characteristics of the diode were analyzed using curve fitting techniques to extract the ideality factor (η) and saturation current (I_s) parameters. We used schockley equation for fitting :

$$I = I_s \left(e^{\frac{V}{\eta V_T}} - 1 \right) \tag{1}$$

where $V_T = \frac{kT}{q} = 25.85$ mV at room temperature is the thermal voltage.

Three different series resistances were used in the measurements: 100 Ω, 220 Ω, and 4700 Ω. For each resistance value, the I-V data was fitted using gnuplot

Gnuplot Script Implementation

The curve fitting was performed using the gnuplot script shown in Listing 1.The thermal voltage V_T is fixed at 25.85 mV, while the saturation current I_s and ideality factor n are fitted parameters. Initial guesses of $I_s = 1 \times 10^{-12}$ A and $n = 1.5$ were used for the fitting process.

Listing 1: Gnuplot script for I-V curve fitting

```
# fit_diode.plt
# Gnuplot script to fit diode IV data

f(V) = Is*(exp(V/(n*Vt)) - 1)

Vt = 0.02585     # thermal voltage at room temp (fixed)
Is = 1e-12       # initial guess for saturation current
n  = 1.5         # initial guess for ideality factor

fit f(x) 'diode_IV_4700.dat' using 1:2 via Is, n #change the .dat
    file accordingly for different resistances

set xlabel "V_{diode} (V)"
set ylabel "I (A)"
set grid

set terminal pngcairo size 800,600 enhanced font 'Arial,12'
set output '4700fit.png'

plot 'diode_IV_4700.dat' using 1:2 title 'Data' with linespoints
   pt 7, \
     f(x) title 'Fitted curve' with lines lw 2
```

The complete gnuplot log file with detailed fitting results and convergence information can be found here

Extracted Parameters

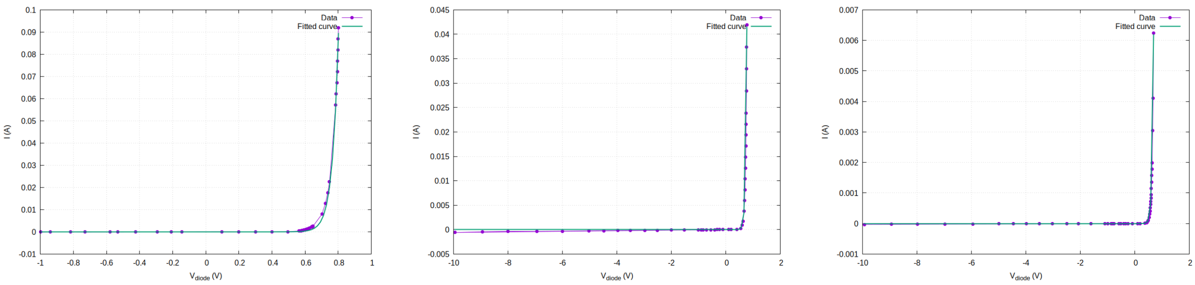
The fitted parameters for the three different series resistance configurations are summarized in Table 1.

Table 1: Extracted diode parameters for different series resistances

Series Resistance (Ω)	**Ideality Factor** η	**Saturation Current** I_s **(A)**	**Relative Error in** I_s **(%)**
100	1.405 ± 0.028	$2.309 \times 10^{-11} \pm 9.88 \times 10^{-12}$	42.8
220	1.673 ± 0.014	$7.035 \times 10^{-10} \pm 9.92 \times 10^{-11}$	14.1
4700	1.855 ± 0.005	$3.491 \times 10^{-9} \pm 1.33 \times 10^{-10}$	3.8

I-V Characteristic Curves

The experimental data points and fitted curves for all three resistance configurations are shown in Figure .



(a) R = 100 Ω (b) R = 220 Ω (c) R = 4700 Ω

I-V characteristics and fitted curves for diode with different series resistances.

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